UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/261,598	07/14/2023	Lisa DEROSA	B23-5214QTUS	7080

76392 7590 02/23/2026

ARRIGO, LEE, GUTTMAN & MOUTA-BELLUM LLP
1050 30th Street NW
Washington, DC 20007

EXAMINER
SHAHNAN SHAH, KHATOL S

ART UNIT	PAPER NUMBER
1645	

NOTIFICATION DATE	DELIVERY MODE
02/23/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

SAL@ARRIGO.US
contact@arrigo.us
eofficeaction@appcoll.com

Office Action Summary	Application No. 18/261,598	Applicant(s) DEROSA et al.	
	Examiner KHATOL S SHAHNAN SHAH	Art Unit 1645	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 10/28/2025.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1,3,5-12 and 18-27 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1,3,5-12 and 18-27 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☑ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 7/14/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 7/14/2023.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

2. Applicants amended and response of 10/28/2025 are acknowledged. Claims 1, 3 and 12 have been amended. Clams 2, 4 and 13-17 have been canceled. New clams 18-27 have been added.

Status of Claims

3. Claims 1, 3, 5-12 and 18-27 are pending in this application. Claims 1, 3 and 12 have been amended. Clams 2, 4 and 13-17 have been canceled. New clams 18-27 have been added.

Specification

5. Specification has been objected by the examiner. Pages 50-55 contain sequences that are not listed in 1449.

KHATOL, SEQUENCES ARE NOT LISTED ON A 1449. PLEASE CORRECT.

Drawings

6. The drawings submitted by the applicant dated 7/14/2023 have been accepted by the examiner.

Information Disclosure Statement

7. Applicants' information disclosure statement filed 7/14/2023 has been considered. Initialed copy of 1449 is enclosed. However, the listing of references in the specification pages 50-55 is not a proper information disclosure statement. 37 CFR 1.98(b) requires a list of all patents, publications, or other information submitted for consideration by the Office, and MPEP § 609.04(a) states, "the list may not be incorporated into the specification but must be submitted in a separate paper." Therefore, unless the references have been cited by the examiner on form PTO-892, they have not been considered.

Response to Restriction Requirement

8. Applicant's election of 10/28/2025 is acknowledged. Applicants elected group I claims 1,3 and 5-12 drawn to method of determining if a cancer patient is likely to be a

good responder to immune checkpoint inhibitor therapy. New claims 18-27 will be also examined under elected group I. Because applicant did not distinctly and specifically point out the supposed errors in the restriction requirement, the election has been treated as an election without traverse (MPEP § 818.01(a)). Claims 1,3, 5-12 and 18-27 are under consideration.

Claim Objections

9. Claims 8 and 23 are objected to because of the following informalities: The claims recite multiple abbreviations anti-PD1/PD-L1/PD-L2 etc. Full name of said abbreviations are required when appears in the claims for the first time. Appropriate correction is required.

Claim Rejections• 35 USC§ 101

35 U.S.C. 101 reads as follows:

> Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

10. Claims 1,3, 5-12 and 18-27 are rejected under 35 U.S.C. 101 because the claimed invention is directed to a natural phenomenon, specifically natural bacteria, without significantly more.

Claim 1 recite:

Claims 1. An in vitro theranostic method of determining if a cancer patient is likely to be a good responder to an immune checkpoint inhibitor (ICI)-based therapy, comprising measuring, in a sample from said patient, the relative abundance of bacteria of the Akkermansia genus, wherein the presence of bacteria of the Akkermansia genus below a predetermined threshold is indicative that the patient is likely to be a good responder to the ICI-based therapy, and administering the ICI-based therapy to the

cancer patient if the abundance of bacteria of the Akkermansia genus is below the predetermined threshold.

This claim really has two embodiments, each of which must be eligible if we are going to find the claim eligible.

Embodiment 1:

Measure relative abundance of bacteria

Presence of bacteria < threshold

Administer ICI.

This embodiment is eligible. The claim does describe the law of nature, namely that patients with high levels of the bacteria are more likely to be responders. Then this embodiment integrates the law of nature into a practical application. In particular, the claim requires administering an ICI. These are the appropriate treatments for patients with cancer, and in particular those who also have high levels of the bacteria.

However the claim also has

Embodiment 2:

Measure relative abundance of bacteria

Presence of bacteria > threshold

The "administering the ICI-based therapy ... if the abundance of bacteria ... is below the predetermined threshold" at the end of the claim is a conditional limitation.

Because Embodiment #2 above does not integrate the exception into a practical application, the answer to 2A prong 2 is NO.

Then we move to step 2B.

Then only other step is measuring the relative abundance of bacteria of the Akkermansia genus. This is recited at a very high level of generality, and encompasses all possible ways of measuring. It also is a necessary data-gathering step required to feed into the law of nature. This weighs against being "significantly more". This also applies to claims 3 and 18.

The claimed invention is directed to a judicial exception (i.e., a law of nature, a natural phenomenon, or an abstract idea) without significantly more. Claim(s) is/are directed to

natural product. The claim(s) does/do not include additional elements that are sufficient to amount to significantly more than the judicial exception because.

The characteristics of each component are not markedly different from their naturally occurring counterparts since they have the same structure and function as they do when find in nature.

Step 1: Yes, the claim is drawn to natural product, which is one of the four statutory categories.

Step 2A: yes, the claim is drawn to a judicial exception (JE).

Step 2B. This step fails to add significantly more than the JE. Therefore, that would not be enough to make it eligible.

The claimed mixture is like the novel bacterial mixture of Funk Brothers, which was held ineligible because each species of bacteria in the mixture (like each component in the foliar spray mixture) continued to have "the same effect it always had", i.e., it lacked markedly different characteristics. Funk Brothers Seed Co. v. Kala Inoculant Co., 333 U.S. 127, 131 (1948), discussed in Myriad Genetics, 133 S. Ct. at 2117 (explaining that the bacterial mixture of Funk Brothers "was not patent eligible because the patent holder did not alter the bacteria in any way"). While not discussed in the opinion, it is noted that several of the claims held ineligible in Funk Brothers recited specific amounts of the bacterial species in the mixture, e.g., claims 6, 7 and 13. Funk Brothers, 333 U.S. at 128 n.1.

It is well established that the mere physical or tangible nature of additional elements such as adding water or saline does not automatically confer eligibility on a claim directed to an abstract idea {see, e.g., Alice Coro, v. CLS Bank Int'l, 134 S.Ct. 2347, 2358-59 (2014)). Thus, taken alone, the hybridizing step does not amount to significantly more.

Claim Rejections - 35 USC § 112

The following is a quotation of 35 U.S.C. 112(b):

(b) CONCLUSION .— The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:

The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

11. Claims 1,3, 5-12 and 18-27 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention. Claim 1 recites:

Claims 1. An *in vitro* theranostic method of determining if a cancer patient is likely to be a good responder to an immune checkpoint inhibitor (ICI)-based therapy, comprising measuring, in a sample from said patient, the relative abundance of bacteria of the *Akkermansia* genus, wherein the presence of bacteria of the *Akkermansia* genus below a predetermined threshold is indicative that the patient is likely to be a good responder to the ICI-based therapy, and administering the ICI-based therapy to the cancer patient if the abundance of bacteria of the *Akkermansia* genus is below the predetermined threshold.

The claim missing a correlation step. The claim determining if a cancer patient is likely to be a good responder to an immune checkpoint inhibitor (ICI)-based therapy. The claim recites relative abundance of bacteria and predetermined threshold. How are these steps are determined? Are other steps missing from the method.

Additionally, claims 1 and 18 are unclear how define “a good responder”. The metes and bounds for determining a good responder are unclear. The claim does not recite aa comparison to determine whether a patient is a good responder. Therefore, clarification is required to overcome the rejection.

Claim Rejections - 35 USC § 112

12. The following is a quotation of the first paragraph of 35 U.S.C. 112(a):

(a) IN GENERAL.—The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise,

and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention.

The following is a quotation of the first paragraph of pre-AIA 35 U.S.C. 112:

The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor of carrying out his invention.

13. Claims 5, 19, 20 are rejected under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-AIA), first paragraph, as failing to comply with the enablement requirement. The claim(s) contains subject matter which was not described in the specification in such a way as to enable one skilled in the art to which it pertains, or with which it is most nearly connected, to make and/or use the invention. This is deposit rejection.

The invention appears to employ novel strains of bacteria. It is not clear if the written description is sufficiently repeatable to avoid the need for a deposit. Further it is unclear if the starting materials were readily available to the public at the time of invention.

It appears that a deposit was made in this application as filed as noted on pages 7 and 8 of the specification. However, it is not clear if the deposits meet all the criteria set forth in 37 CFR 1.801-1.809. Because it is not clear that organisms having the accession numbers of Akkermansia SGB9228 and Akkermansia muciniphila are known and publicly available or can be reproducibly isolated from nature without undue experimentation. Without a publicly available deposit of the above strains, one of ordinary skill in the art could not be assured of the ability to practice the invention as claimed. Exact replication of the strains is an unpredictable event.

Applicant's referral to the deposit of strains of Akkermansia SGB9228 and Akkermansia muciniphila on paragraphs 0081, 0095, 0096 and 0157 of the specification is an insufficient assurance that all required deposits have been made and all the conditions of 37 CFR 1.801-1.809 have been met. If the deposit has been made under the provisions of the Budapest Treaty, filing of an affidavit or declaration by applicant or assignees or a statement by

an attorney of record who has authority and control over the conditions of deposit over his or her signature and registration number stating that the deposit has been accepted by the International Depository Authority under the provisions of the Budapest Treaty and that all restrictions upon public access to the deposit will be irrevocably removed upon the grant of a patent on this application. These requirements are necessary when deposits are made under the provisions of the Budapest Treaty as the Treaty leaves this specific matter to the discretion of each State. Amendment of the specification to recite the date of the deposit and the complete name and full street address of the depository is required.

If the deposits have not been made under the provisions of the Budapest Treaty, then in order to certify that the deposits comply with the criteria set forth in 37 CFR 1.801-1.809, assurances regarding availability and permanency of deposits are required. Such assurance may be in the form of an affidavit or declaration by applicants or assignees or in the form of a statement by an attorney of record who has the authority and control over the conditions of deposit over his or her signature and registration number averring:

(a) during the pendency of this application, access to the deposits will be afforded to the Commissioner upon request;

(b) all restrictions upon the availability to the public of the deposited biological material will be irrevocably removed upon the granting of a patent on this application;

(c) the deposits will be maintained in the public repository for a period of at least thirty years from the date of deposit or for the enforceable life of the patent of or for a period of five years after the date of the most recent request for the furnishing of a sample of the deposited biological material, whichever is longest; and

(d) the deposits will be replaced if they should become nonviable or non-replicable.

In addition, a deposit of biological material that is capable of self-replication either directly or indirectly must be viable at the time of deposit and during the term of deposit. Viability may be tested by the repository. The test must conclude only that the deposited material is capable of reproduction. A viability statement for each deposit of biological material not made under the Budapest Treaty must be filed in the application and must contain:

1) The name and address of the depository;

2) The name and address of the depositor;

3) The date of deposit;

4) The identity of the deposit and the accession number given by the depository;

5) The date of the viability test;

6) The procedures used to obtain a sample if test is not done by the depository; and

7) A statement that the deposit is capable of reproduction.

As a possible means for completing the record, applicant may submit a copy of the contract with the depository for deposit and maintenance of each deposit.

If the deposit was made after the effective filing date of the application for patent in the United States, a verified statement is required from a person in a position to corroborate that the strains described in the specification as filed is the same as that deposited in the depository. Corroboration may take the form of a showing a chain of custody from applicant to the depository coupled with corroboration that the deposit is identical to the biological material described in the specification and in the applicant's possession at the time the application was filed.

Applicant's attention is directed to In re Lundack, 773 F.2d.1216, 227 USPQ (CAFC 1985) and 37 CFR 1.801-1.809 for further information concerning deposit practice.

Claim Rejections - 35 USC § 103

14. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

15. Claims 1, 3, 5-12 and 18-27 are rejected under 35 U.S.C. 103 as being un-patentable over Zitvogel et al. WO 2018115519 A1 filed 6/28/2018 in view of Dutta et al. Biomarker Research vol. 8, no.1, 8/12/2020 and further in view of Weir et al. PLOS One vol.8, no. 8, 8/6/2013. All art of record search report.

The claims are drawn to:

Claims 1. An *in vitro* theranostic method of determining if a cancer patient is likely to be a good responder to an immune checkpoint inhibitor (ICI)-based therapy, comprising measuring, in a sample from said patient, the relative abundance of bacteria of the *Akkermansia* genus, wherein the presence of bacteria of the *Akkermansia* genus below a predetermined threshold is indicative that the patient is likely to be a good responder to the ICI-based therapy, and administering the ICI-based therapy to the cancer patient if the abundance of bacteria of the *Akkermansia* genus is below the predetermined threshold.

Zitvogel et al. recite the present invention pertains to an *in vitro* theranostic method of determining if a cancer patient is likely to be a good responder to an anti-PD1/PD-L 1/PD-L2 Ab-based therapy, comprising:

(i) assessing, in a feces sample from said patient, the relative abundance of at least 10 microorganism species selected from the microorganism species disclosed in Table 1 and Table 2;

(ii) for each microorganism, comparing the relative abundance measured

in step (i) to a predetermined threshold,

wherein over-representation of microorganism species disclosed in Table 1 (below) and under-representation of microorganism species disclosed in Table 2 (below) are indicative that the patient is likely to be a good responder to the anti-PD1/PD-L1/PD-L2 Ab-based therapy. See page 6.

Zitvogel et al. further recite:

The present invention thus pertains to a theranostic method for determining if a cancer patient needs a bacterial compensation before administration of an anti-PD1/PD-L1/PD-L2 Ab-based therapy comprising assessing, in a feces sample from said patient, the presence or absence of *Akkermansia muciniphila,* wherein if *Akkermansia muciniphila* is absent from said feces sample, the patient needs a bacterial compensation with a bacterial composition or a fecal microbial composition as above-described. See page 44.

Zitvogel et al. teach of independent claims 1, 18 and claim. Zitvogel et al. which discloses an in vitro theranostic method of determining if a cancer patient is likely to be a good responder to an anti-PD1/PD-L1/PD-L2 Ab-based therapy, comprising assessing, in a feces sample from said patient, the relative abundance of microorganisms including Akkermansia muciniphila, and comparing the relative abundance of said organism to a predetermined threshold, wherein over-representation of Akkermansia muciniphila is indicative that the patient is likely to be a good responder to the anti-PD1/PD-L1/PD-L2 Ab-based therapy (abstract and page 6). Zitvogel et al. teach limitations of claims 5, 19 and 20 Akkermansia muciniphila. (see abstract and claims 1-3). Zitvogel et al. teach limitations of claims 3, 5, 9, 24, 25 NSCLC, kidney cancer see pages 3, 9, 23, 24. Zitvogel et al. teach limitations of clam 27 see pages 23, 24. Zitvogel et al. discloses a bacterial composition comprising Akkermansia muciniphila, for use in the treatment of a cancer patient having no Akkermansia muciniphila in his/her intestinal microbiota, wherein said composition is used in combination with an ICI-based therapy (Zitvogel et al., page 44, lines 31-37). Zitvogel et al. teach FMT see pages 50-51.

The difference between the claimed subject-matter and Zitvogel et al. is that in addition, patients with an abundance of A. muciniphila above a higher predetermined

threshold, are poor responders to anti-PD1/ PD-L1/PD-L2 antibody-based therapy. The effect is that poor responders with increased A. muciniphila may be identified.

Therefore, Dutta et al. discloses that overgrowth or over-abundance of A. muciniphila may be detrimental. Antibiotic-mediated reduction in gut microbiota diversity and enrichment of A. muciniphila promoted inflammation in mammary tumors and increased metastatic dissemination of tumor cells. Dutta et al states that a threshold dose needed for A. muciniphila to enhance cancer treatment response without inducing negative effects will need to be established before any isolated supplementation or enrichment is attempted to improve the outcome of cancer therapy (Dutta et al 2, page 11). Dutta et al therefore clearly states that there are a maximum levels of A. muciniphila, which must not be exceeded (Dutta et al. page 11, left column).

Weir et al. disclose that the presence of A. muciniphila above a threshold is indicative of dysbiosis, and hence, presence of this microorganism below this threshold is indicative that there is no dysbiosis (Weir et al. abstract; page 7).

Therefore, it would have been *prima facie* obvious at the time of applicants' invention to combine the methods and compositions of references to obtain the instant invention. It would have been obvious to one of ordinary skill in the art that the method of Zitvogel et al. which teach an *in vitro* theranostic method of determining if a cancer patient is likely to be a good responder to an anti-PD1/PD-L 1/PD-L2 Ab-based therapy and Weir et al. disclose that the presence of A. muciniphila above a threshold is indicative of dysbiosis, and hence, presence of this microorganism below this threshold is indicative that there is no dysbiosis. And the benefit of using Dutta et al is that because Dutta et al states that a threshold dose needed for A. muciniphila to enhance cancer treatment response without inducing negative effects will need to be established before any isolated supplementation or enrichment is attempted to improve the outcome of cancer therapy.

Additionally, *KSR International Co. v. Teleflex Inc.,* 127 S. Ct. 1727, 1741 (2007), discloses combining prior art elements according to known methods to yield predictable results, thus the combination is obvious unless its application is beyond that person's

skill. *KSR International Co. v. Teleflex Inc.,* 127 S. Ct. 1727, 1741 (2007) also discloses that "The combination of familiar element according to known methods is likely to be obvious when it does no more than yield predictable results". It is well known to combine known methods which function in a predictable manner to yield a reasonable expectation of success along with predictable results to one of ordinary skill in the art at the time of the invention. Thus, it would have been obvious to a person of ordinary skill in the art to combine prior art elements according to known methods that is ready for improvement to yield predictable results. The claimed invention is *prima facie* obvious in view of the teachings of the prior art, absent any convincing evidence to the contrary.

Conclusion

16. No claims are allowed.

17. Any inquiry concerning this communication or earlier communications from the examiner should be directed to KHATOL S SHAHNAN SHAH whose telephone number is (571)272-0863. The examiner can normally be reached on Mon-Tue, Thurs-Fri 12pm-8pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Daniel Kolker can be reached on 571-272-8131. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see https://ppair-my.uspto.gov/pair/PrivatePair. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access

to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/KHATOL S SHAHNAN SHAH/

Examiner, Art Unit 1645

February 7, 2026

/JANA A HINES/
Primary Examiner, Art Unit 1645